

Aditya Kulkarni

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PROFESSIONAL SUMMARY

MS CS candidate (NC State, May 2026) specializing in high-performance distributed systems and production software engineering. Built event-driven infrastructure handling **2,000 concurrent inference requests/second** with zero message loss using Temporal.io and Kafka. Engineered kernel-level virtual memory systems in C (Xinu OS, 25% page fault reduction) and reduced processing latency 20% on real-time data pipelines with Hadoop and Spark. Strong quantitative foundation in statistics, probability, and ML -- with hands-on finance domain experience building automated data systems. Ready to apply systems engineering depth to automated trading infrastructure.

CORE COMPETENCIES

High-Performance Systems

C++ & Python Engineering

Distributed Systems & Kafka

Low-Latency Architecture

Event-Driven Design

Quantitative Analysis

Production Software Delivery

Linux & Systems Programming

WORK EXPERIENCE

Isomer AI May 2025 - Present

Product Engineer Intern

- Designed and delivered **10+ production-quality components** for a live automation platform -- Temporal.io workflow orchestration, real-time event pipelines, RBAC, document processing (PDF, email, S3) -- across 7 deployments and 394 commits
- Engineered low-latency backend services in TypeScript/Node.js with PostgreSQL and Redis; integrated Sentry observability and monitored system health across production workloads

Capgemini Jun 2023 - Aug 2023

AI/ML Intern

- Engineered distributed data pipelines using **Hadoop + Spark** for batch processing and integrated Kafka streaming for real-time data ingestion -- reduced ticket-processing latency by **20%** and end-to-end processing time by 15%

Mphasis (HP Company) Jun 2024 - Aug 2024

Software Developer Intern

- Built 5+ automation workflows including ML-driven claim processing pipelines (OCR, NLP); automated deployments on AWS and GCP with Grafana monitoring dashboards and model validation

NC State University Apr 2025 - Present

Graduate Research Assistant

- Designed 500GB cloud data platform on GCP (100+ researchers); applied hyperparameter tuning and time-series analysis to improve ML model accuracy by 12%

PROJECTS

AlgoPulse -- High-Performance Agentic Platform Independent, 2025

Identified API-polling bottleneck under load and re-architected to event-driven streaming. Scaled worker architecture with **Temporal.io state machines and Kafka** for fault-tolerant message brokering -- sustained **2,000 async requests/second** with zero message loss under concurrent load.

TypeScript · Python · Temporal.io · Kafka · Supabase Edge Functions · Node.js

Demand Paging -- Xinu OS Kernel Systems Programming, 2024

Engineered an 8-store virtual memory system in **C for Xinu OS** with Second-Chance page replacement algorithm; reduced page faults by **25%**. Added 7+ kernel-level system calls using inverted page tables and a custom fault handler, improving memory performance by 15%.

C · Xinu OS · Virtual Memory · Kernel Systems · Inverted Page Tables

PerCent -- Automated Finance Platform HackNC24 Top 10 / 100+ teams

Built a full finance automation platform with goal tracking, expense logging, and an AI-driven data pipeline for financial domain reasoning -- **30% reduction in manual research time**. Top 10 of 100+ teams in 48 hours.

Python · Hugging Face · Qwen-14B · RAG · Selenium

LLM Eval Harness -- Quantitative Model Benchmarking Independent, 2025

Built a vendor-neutral benchmark comparing 3 LLM providers across 2,000 examples with statistical rigor -- paired bootstrap CIs, McNemar test, Holm correction. Automated cost tracking (per-call and per-experiment) and latency analysis (p50/p95) across async evaluation runs.

Python · DuckDB · Parquet · Streamlit · Async · Statistical Analysis

EDUCATION

MS Computer Science NC State University Aug 2024 - May 2026

Algorithms, Operating Systems, Data Science, Artificial Intelligence, Software Engineering

B.Tech Computer Science Mumbai University Aug 2020 - May 2024

CGPA 9.32/10 · Machine Learning, Big Data Infrastructure, DS & Algorithms, Statistics, Blockchain

SKILLS

Languages: C, C++ (coursework), Python, TypeScript, JavaScript, Java, SQL, R

Systems & Infra: Temporal.io, Kafka, Spark, Hadoop, Linux, Docker, Kubernetes, AWS, GCP, CI/CD, Grafana, Sentry

Frameworks: Node.js, React, PostgreSQL, Redis, Prisma, Express, Distributed Systems

Quantitative: Statistics, Probability, ML (TensorFlow, PyTorch), NumPy, Pandas, Time-Series Analysis

Awards: Top 10 in 6+ national hackathons · Google Developer Student Club Lead (500+ students) · IEEE Published